REFINING A HYPERCOVERING ALONG A COVERING OF ONE LEVEL

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1631-1642 of sources/stacks-hypercovering/pinned-original-source.tex; proof span is lines 1644-1773, the complete author proof (128 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 1 of queue_hypercovering.json (pb=4650, stmt 1631-1642, proof 1644-1773).

Editorial note. stacks-hypercovering wave-1 (hc01): fresh adjudication accept (2026-10-14); pool per standing fallback order (hypercovering -> pic -> spaces-more-cohomology) after md01 closed the moduli pool; hypercovering.tex 53 labels/34 with proof judged 26 accept / 8 reject / 14 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed. Named omission: Author omits the computation identifying the map S' -> S and the final combinatorial lifting claim ('easy to see and omitted') - named mechanical checks; the construction and covering arguments are in-span.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 01G4: sources/stacks-hypercovering/pinned-original-source.tex, source lines 524–535. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01GH: sources/stacks-hypercovering/pinned-original-source.tex, source lines 1511–1536. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0182: sources/stacks-hypercovering/pinned-original-source.tex, source lines 1773–1773. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0B13: sources/stacks-hypercovering/pinned-original-source.tex, source lines 1773–1773. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0186: sources/stacks-hypercovering/pinned-original-source.tex, source lines 1773–1773. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 017M: sources/stacks-hypercovering/pinned-original-source.tex, source lines 1773–1773. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0167: sources/stacks-hypercovering/pinned-original-source.tex, source lines 1773–1773. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $\mathcal{C}$ be a site with fibre products. Let X be an object of $\mathcal{C}$. Let K be a hypercovering of X. Let $k \geq 0$ be an integer. Let $u : Z \to K_k$ be a covering in $SR(\mathcal{C}, X)$. Then there exists a morphism of hypercoverings $f : L \to K$ such that $L_k \to K_k$ factors through u.*

Proof. Denote $Y = K_k$. Let $C[k]$ be the cosimplicial set defined in Simplicial, Example 0B13. We will use the description of $\mathrm{Hom}(C[k], Y)$ and $\mathrm{Hom}(C[k], Z)$ given in Simplicial, Lemma 017M. There is a canonical morphism $K \to \mathrm{Hom}(C[k], Y)$ corresponding to $\mathrm{id} : K_k = Y \to Y$. Consider the morphism $\mathrm{Hom}(C[k], Z) \to \mathrm{Hom}(C[k], Y)$ which on degree n terms is the morphism

$$\prod\nolimits_{\alpha:[k]\to[n]} Z \longrightarrow \prod\nolimits_{\alpha:[k]\to[n]} Y$$

using the given morphism $Z \to Y$ on each factor. Set

$$L = K \times_{\mathrm{Hom}(C[k],Y)} \mathrm{Hom}(C[k], Z).$$

The morphism $L_k \to K_k$ sits in to a commutative diagram

$$\begin{array}{ccccc} L_k & \longrightarrow & \prod_{\alpha:[k]\to[k]} Z & \xrightarrow{\mathrm{pr}_{\mathrm{id}_{[k]}}} & Z \\ \downarrow & & \downarrow & & \downarrow \\ K_k & \longrightarrow & \prod_{\alpha:[k]\to[k]} Y & \xrightarrow{\mathrm{pr}_{\mathrm{id}_{[k]}}} & Y \end{array}$$

Since the composition of the two bottom arrows is the identity we conclude that we have the desired factorization.

We still have to show that L is a hypercovering of X. To see this we will use Lemma 01GH. Condition (1) is satisfied by assumption. For (2), the morphism

$$\mathrm{Hom}(C[k], Z)_0 \to \mathrm{Hom}(C[k], Y)_0$$

is a covering because it is isomorphic to $Z \to Y$ as there is only one morphism $[k] \to [0]$.

Let us consider condition (3) for $n = 0$. Then, since $(\mathrm{cosk}_0 T)_1 = T \times T$ (Simplicial, Example 0182) and since $\mathrm{Hom}(C[k], Z)_1 = \prod_{\alpha:[k]\to[1]} Z$ we obtain the diagram

$$\begin{array}{ccc} \prod_{\alpha:[k]\to[1]} Z & \longrightarrow & Z \times Z \\ \downarrow & & \downarrow \\ \prod_{\alpha:[k]\to[1]} Y & \longrightarrow & Y \times Y \end{array}$$

with horizontal arrows corresponding to the projection onto the factors corresponding to the two nonsurjective α. Thus the arrow γ is the morphism

$$\prod\nolimits_{\alpha:[k]\to[1]} Z \longrightarrow \prod\nolimits_{\alpha:[k]\to[1] \text{ not onto}} Z \times \prod\nolimits_{\alpha:[k]\to[1] \text{ onto}} Y$$

which is a product of coverings and hence a covering by Lemma 01G4.

Let us consider condition (3) for $n > 0$. We claim there is an injective map $\tau : S' \to S$ of finite sets, such that for any object T of $\mathrm{SR}(\mathcal{C}, X)$ the morphism

$$\mathrm{Hom}(C[k], T)_{n+1} \to (\mathrm{cosk}_n \mathrm{sk}_n \mathrm{Hom}(C[k], T))_{n+1} \tag{0.1.1}$$

is isomorphic to the projection $\prod_{s\in S} T \to \prod_{s'\in S'} T$ functorially in T. If this is true, then we see, arguing as in the previous paragraph, that the arrow γ is the morphism

$$\prod\nolimits_{s\in S} Z \longrightarrow \prod\nolimits_{s\in S'} Z \times \prod\nolimits_{s\notin\tau(S')} Y$$

which is a product of coverings and hence a covering by Lemma 01G4. By construction, we have $\mathrm{Hom}(C[k], T)_{n+1} = \prod_{\alpha:[k]\to[n+1]} T$ (see Simplicial, Lemma 017M). Correspondingly we take $S = \mathrm{Map}([k], [n+1])$. On the other hand, Simplicial, Lemma 0186, provides a description of points of $(\mathrm{cosk}_n \mathrm{sk}_n \mathrm{Hom}(C[k], T))_{n+1}$ as sequences $(f_0, \ldots, f_{n+1})$ of points of $\mathrm{Hom}(C[k], T)_n$ satisfying $d^n_{j-1} f_i = d^n_i f_j$ for $0 \leq i < j \leq n+1$. We can write $f_i = (f_{i,\alpha})$ with $f_{i,\alpha}$ a point of T and $\alpha \in \mathrm{Map}([k], [n])$. The conditions translate into

$$f_{i,\delta^n_{j-1}\circ\beta} = f_{j,\delta^n_i\circ\beta}$$

for any $0 \leq i < j \leq n+1$ and $\beta : [k] \to [n-1]$. Thus we see that

$$S' = \{0, \ldots, n+1\} \times \mathrm{Map}([k], [n]) / \sim$$

where the equivalence relation is generated by the equivalences

$$(i, \delta^n_{j-1} \circ \beta) \sim (j, \delta^n_i \circ \beta)$$

for $0 \leq i < j \leq n+1$ and $\beta : [k] \to [n-1]$. A computation (omitted) shows that the morphism (0.1.1) corresponds to the map $S' \to S$ which sends (i, α) to $\delta^{n+1}_i \circ \alpha \in S$. (It may be a comfort to the reader to see that this map is well defined by part (1) of Simplicial, Lemma 0167.) To finish the proof it suffices to show that if $\alpha, \alpha' : [k] \to [n]$ and $0 \leq i < j \leq n+1$ are such that

$$\delta^{n+1}_i \circ \alpha = \delta^{n+1}_j \circ \alpha'$$

then we have $\alpha = \delta^n_{j-1} \circ \beta$ and $\alpha' = \delta^n_i \circ \beta$ for some $\beta : [k] \to [n-1]$. This is easy to see and omitted. □